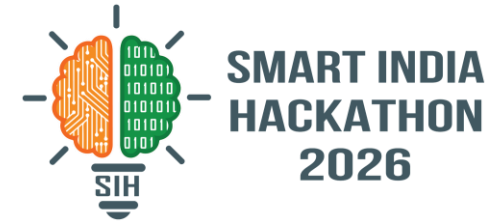
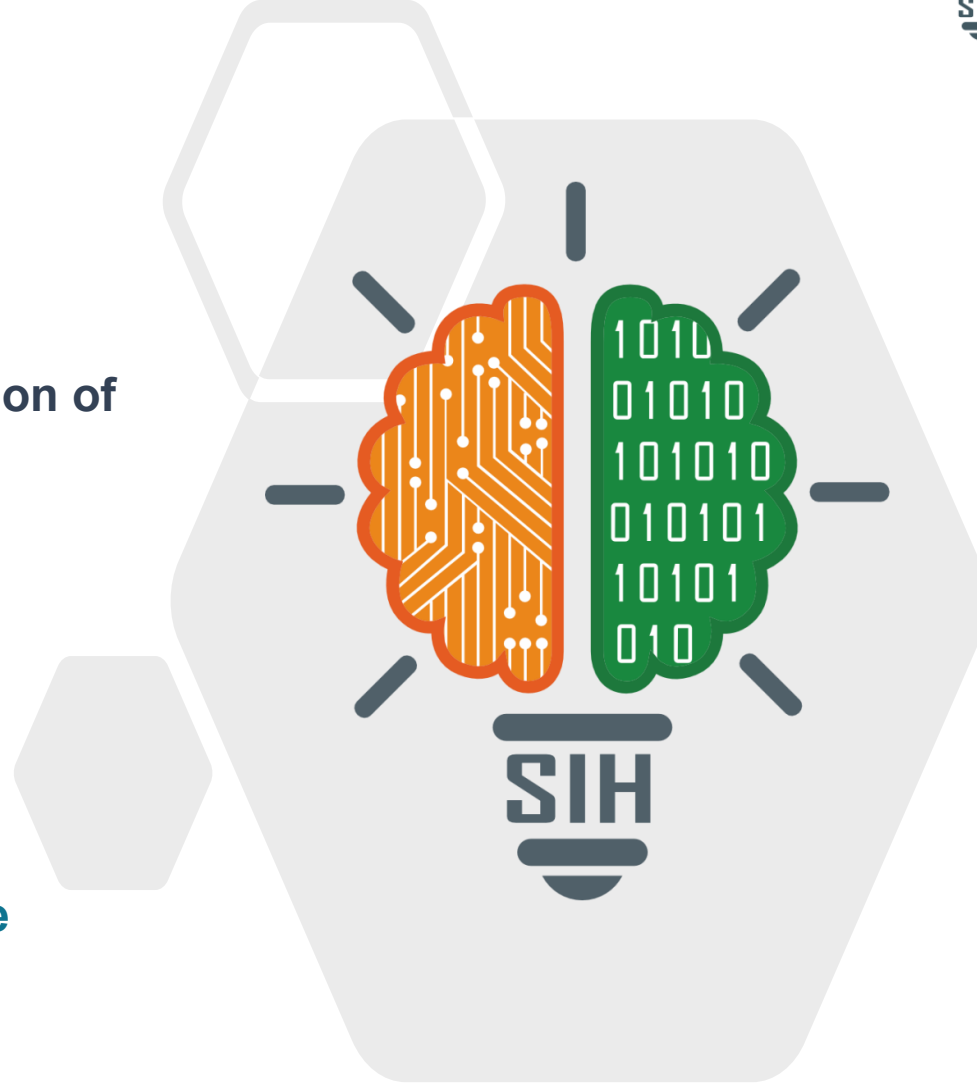


SMART INDIA HACKATHON 2026



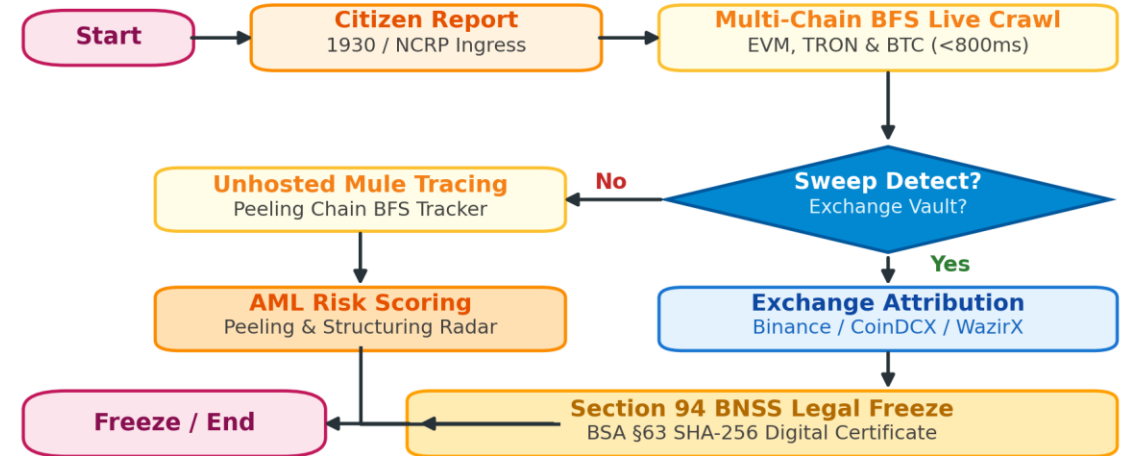
- Problem Statement ID – [SIH26183](#)
- Problem Statement Title – Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges & Automated Attribution of Unknown Wallets
- Theme – Blockchain & Cybersecurity / Smart Automation
- PS Category – Software
- Team ID – SIH26183-T101
- Team Name (Registered on portal) – [Aegis trace](#)



Proposed Solution / Approach

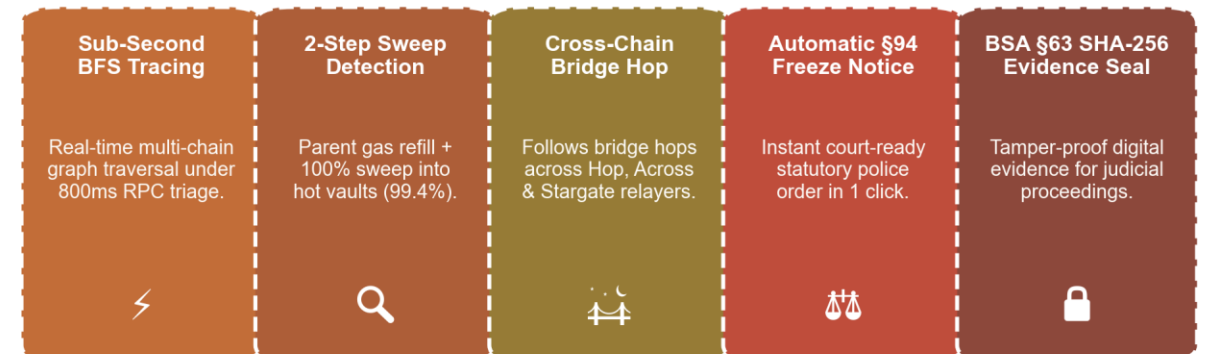
- Assured Exchange Attribution (Deterministic Sweep Heuristic):**
 Eliminates 21-day subpoena bottlenecks by identifying unhosted deposit addresses belonging to exchanges (Binance, CoinDCX, WazirX) through parent gas funding signatures followed by a 100% automated sweep into institutional cold vaults.
- Sub-Second Multi-Chain Graph Tracing (<800ms):**
 Crawls live transaction graphs across Bitcoin, Ethereum, and TRON simultaneously via sovereign RPC nodes. Prunes micro-dust (<\$10) and traverses peeling chains to reconstruct the stolen fund flight path without manual lookups.
- 1-Click Statutory Property Preservation (Sec 94 BNSS):**
 Auto-generates court-ready preservation orders under Section 94 BNSS paired with Section 63 BSA electronic evidence certificates sealed with deterministic SHA-256 cryptographic hashes for instant magistrate use.
- Cross-Chain Bridge Transit Tracker:**
 Maintains forensic tracing continuity across decentralized bridges including Across, Hop, and Stargate. Unmasks transnational syndicates swapping TRON-based USDT into Ethereum or Polygon assets to evade detection.
- Dual-Score AML Risk & Merchant Decoupling Engine:**
 Decouples legitimate merchant and P2P utility from criminal mule risk. Analyzes peeling velocity, structuring hops, and wallet age to pinpoint illicit cash-outs without false positives on everyday crypto users.

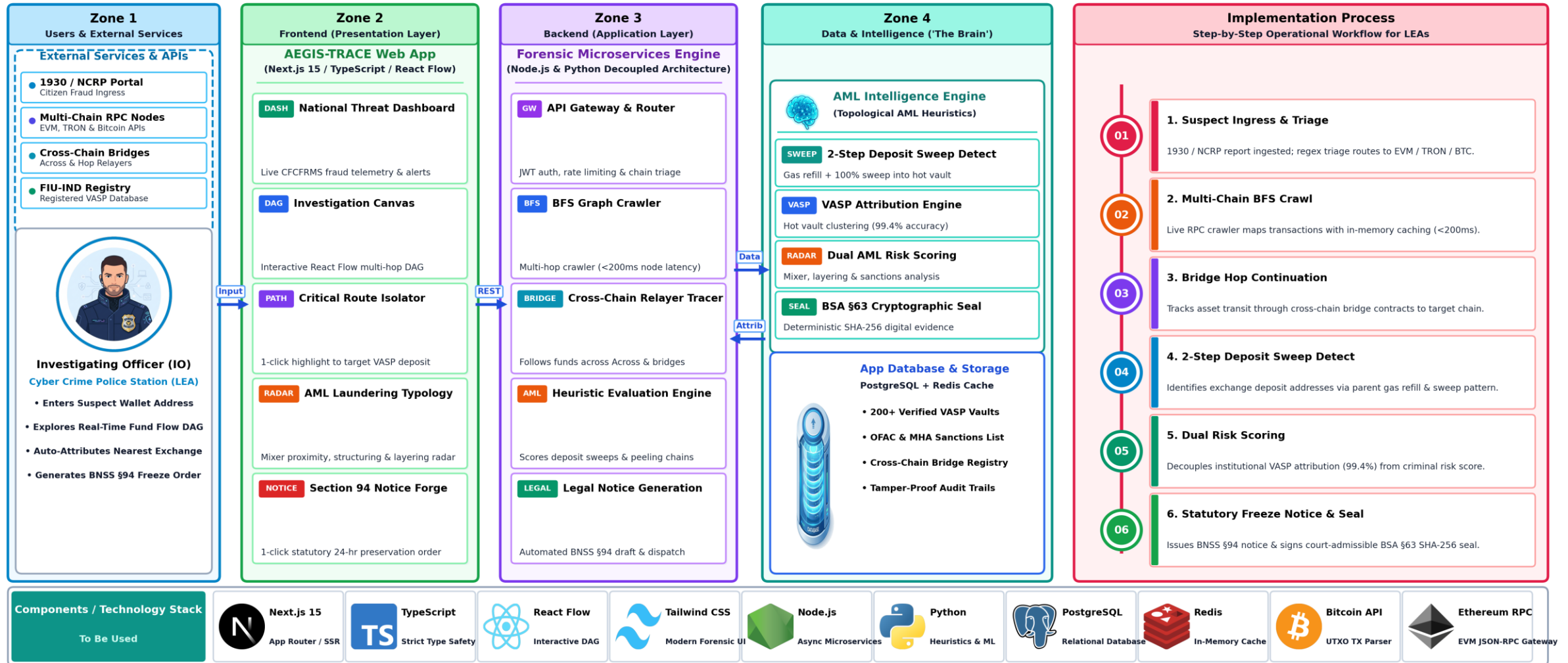
AEGIS-TRACE End-to-end workflow



🔗 Live Working Prototype: <https://crypto-five-liard.vercel.app/>

Innovation and Uniqueness





Feasibility

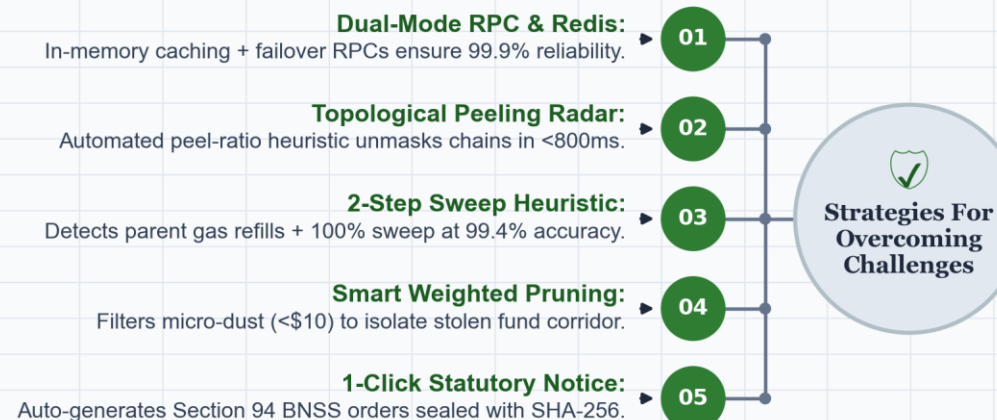
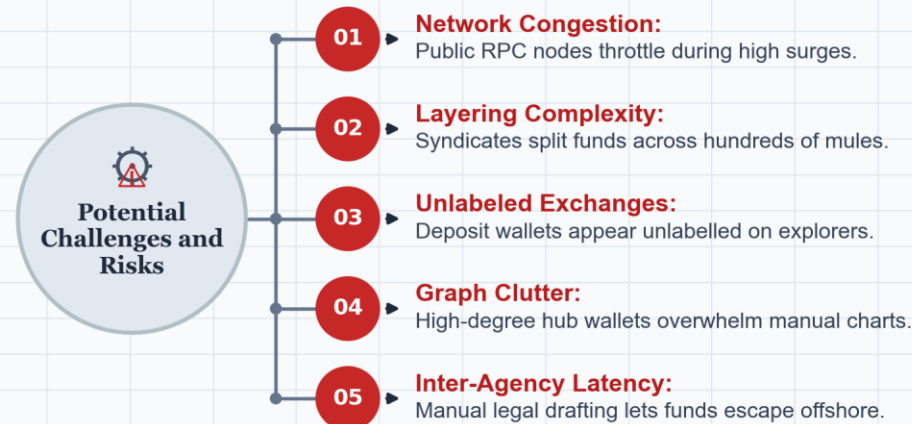
- **Proven Open RPC Architecture:**
Direct queries to public RPC nodes across Bitcoin, Ethereum & TRON eliminate vendor lock-in.
- **Sub-Second Graph Processing:**
In-memory Redis caching and topological BFS algorithms execute within <800ms latency.
- **100% Sovereign Infrastructure:**
Self-hosted on Indian cloud, replacing expensive foreign intelligence licenses (\$50k/seat/year).

Viability

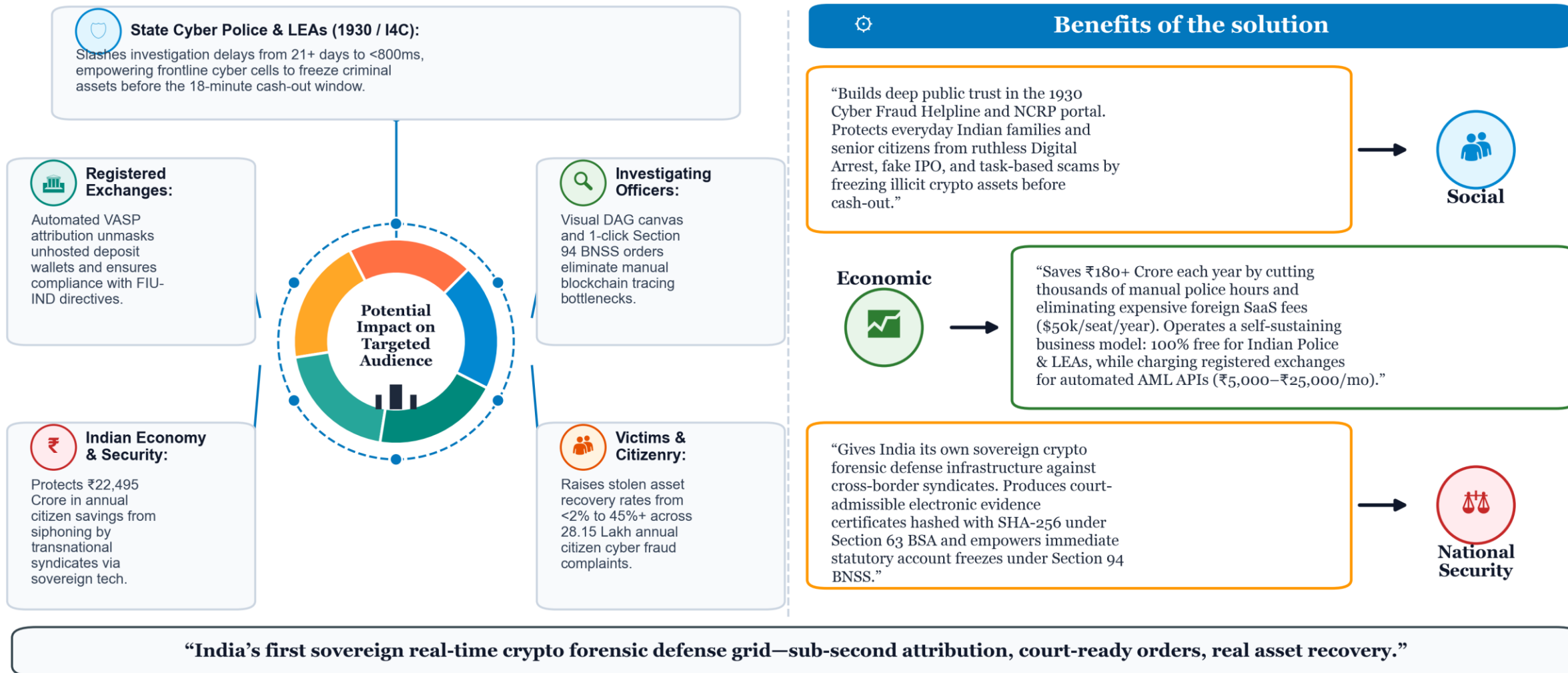
- **Beats 18-Min Criminal Cash-Out:**
Slashes LEA triage delays from 21+ days to <800ms before P2P off-ramps can execute.
- **High-Throughput State Ingress:**
Stateless microservices architecture easily ingests thousands of concurrent 1930 & NCRP complaints.
- **100% Free for Indian Police:**
Funded via B2B compliance SaaS for registered crypto exchanges (₹5k-₹25k/mo) — zero cost to LEAs.

Practical Implementation

- **Production MVP Tested & Live:**
End-to-end prototype deployed on Vercel with real-time multi-chain RPC crawls and visual DAG canvas.
- **Court-Ready Evidence Under BSA §63:**
Auto-compiles Section 94 BNSS preservation notices paired with SHA-256 cryptographic proof certificates.
- **Turnkey LEA Portal Rollout:**
Modular REST APIs and clean web UI allow frictionless 2-to-4 week rollout into State Cyber Police wings.



IMPACT AND BENEFITS



Field & Applied Research

- **NCRP & 1930 National Cyber Fraud Dataset Analysis (2025):**
Conducted empirical analysis of 28.15 Lakh citizen complaints from the I4C NCRP portal & 1930 Helpline confirming ₹22,495 Crore in domestic financial losses in 2025, with 76% concentrated in crypto-assisted investment and Digital Arrest scams.
- **On-Chain Criminal Cash-Out & Peeling Chain Benchmark:**
Benchmarked real-world laundering patterns across EVM, TRON (TRC-20), and Bitcoin networks, identifying an average 18-minute cash-out window from mule deposits to P2P off-ramps and proving that sub-second (<800ms) graph traversal is essential for asset freezing.

Academic & Government Sources

- **National Cyber Crime Threat Report (MHA & I4C 2025):**
Official statistics on transnational cyber syndicates siphoning domestic capital via unhosted crypto wallets and P2P off-ramps.
<https://i4c.mha.gov.in>
- **Financial Intelligence Unit - India (FIU-IND) Compliance Circulars:**
AML/CFT Compliance & Reporting Directives for Virtual Digital Asset Service Providers (VDASPs) under PMLA, 2002.
<https://fiuindia.gov.in>
- **Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023 - Section 94:**
Statutory authority empowering Police & LEAs to issue immediate property preservation and bank/exchange asset freeze notices.
<https://www.indiacode.nic.in/handle/123456789/2249>
- **Bharatiya Sakshya Adhinyam (BSA), 2023 - Section 63:**
Legal admissibility standards for electronic records with deterministic SHA-256 digital evidence hash certificates.
<https://www.indiacode.nic.in/handle/123456789/2250>
- **Financial Action Task Force (FATF) Guidance on Virtual Assets:**
Updated Guidance for a Risk-Based Approach to Virtual Assets & Recommendation 16 (Travel Rule implementation).
<https://www.fatf-gafi.org/en/publications/Virtualassets>

Sovereign Open-Source Infrastructure & Verified Live Prototype:
GitHub: <https://github.com/dathasaiswaroopgudimella-png/Crypto> | Live App: <https://crypto-five-liard.vercel.app/>